

Kyle Choi

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PROJECT EXPERIENCE

Shorelife: California Beach Health Forecasting (iPhone app, live on the App Store, [code](#)) Apr 2026 – Present

- Built and launched, on my own, a free iPhone app that predicts whether a California beach is safe to swim
- Trained the forecast model on ocean, weather, and stormwater data; it flags 83% of unsafe days, 13% false alarms
- Outperforms the "same as yesterday" forecast by a wide margin on beaches the model has never seen (0.96 vs 0.82)
- Runs itself: the model has retrained and republished its forecast automatically 297 times since launch
- Improved the app with feedback from the UCSD surf club and a San Diego city stormwater expert

XOXO: Couples Mood App with AI-Drawn Faces (iPhone app, v1.0 in App Review) Aug 2026 – Present

- Built an iPhone app where partners share their mood as an AI-drawn face, generated privately on the phone
- Designed a test system that scores whether each face shows the right mood without changing who it looks like
- Tested across 4 iPhones and 2 iOS versions; the face stayed recognizable for all 16 people in the test set
- Found that a shorter prompt halved duplicate images and a wording change fixed all 3 invented features

Senior Capstone with Prism Data (fintech): Bank Transaction Intelligence and Credit Risk Sep 2025 – Mar 2026

- Built an AI model from scratch that sorts bank transactions into spending categories with 95% accuracy
- Built a credit-risk score from income and spending that ranks risky vs. safe borrowers correctly 86% of the time
- Set up a team of AI agents on a local machine to propose and test new data features, cutting manual work 50%
- Kept every customer record on the local machine, never sending data to an outside service

AI Evaluation Research: [How AI Assistants Follow Bad Ideas](#) (Baduk / Go) May 2026 – Jun 2026

- Studied 1,000+ professional Go games to test whether one player's move predicts the opponent's reply
- Found the previous move explains 44% of the reply and lands within three points of it 56% of the time
- Showed an AI assistant kept endorsing a flawed idea until told to answer as a well-known scholar in the field

Cross Market Lag Research (open source: github.com/kylechoi101/cross-market-lag-research) May 2026

- Tested a trading idea I had as a freshman: does one country's stock market predict another's hours later?
- Scanned 13 global markets and 156 market pairs with statistical tests built to catch lucky-looking results
- Found real links between 60 market pairs but too weak to trade on; published the negative result openly

MLB Swing Zone Visualization Apr 2025 – Jun 2025

- Visualized 100+ MLB batters' swing decisions, showing their swing zone grows 35% as the strike count rises
- Built an interactive 3D web visualization and presented it to an audience of 30+ non-technical viewers

WORK EXPERIENCE

UCSD, San Diego California – Research Assistant Jun 2024 – Feb 2025

- Joined a physics lab building a neutrino detector; learned its machine-learning tools alongside PhD researchers
- Split detector data processing across many machines at the San Diego Supercomputer Center

Army, Republic of Korea – Sergeant Feb 2021 – Aug 2022

- Led a unit of 25+ soldiers; ran artillery operations and coordinated with first responders during Covid-19

EDUCATION

University of California San Diego — B.S. Data Science, Jun 2026

- **Major GPA:** 3.8 / 4.0; Cumulative GPA: 3.7 / 4.0
- **Coursework:** Time-Series Analysis, Probabilistic Modeling, Representation Learning, Financial Mathematics
- **Awards:** HDSI Best Visualization Project, Rady Advanced Fundamental Equity Valuation Tournament

Chartered Financial Analyst Exam, CFA® Institute — Passed Levels I and II; Level III result pending

SKILLS

- **Programming:** Python, SQL, Swift, JavaScript, R
- **Machine learning:** PyTorch, XGBoost, scikit-learn, transformers, forecasting and time-series modeling, NLP
- **AI / LLM:** evaluation harness design, multi-agent workflows, local and on-device models, Apple Foundation Models
- **Apps and infrastructure:** FastAPI, Expo, SwiftUI, CloudKit, GitHub Actions CI/CD, Render, AWS, Firebase, Docker
- **Spoken:** Korean, English, Mandarin Chinese